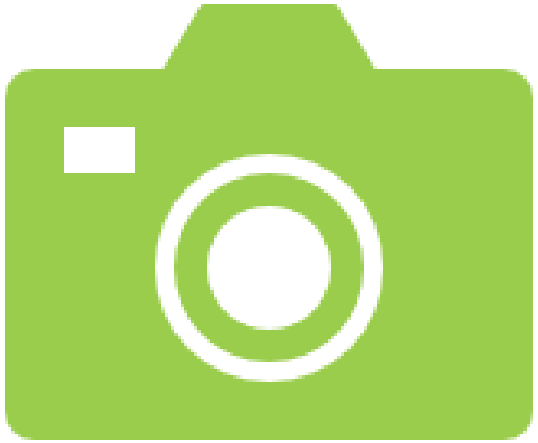
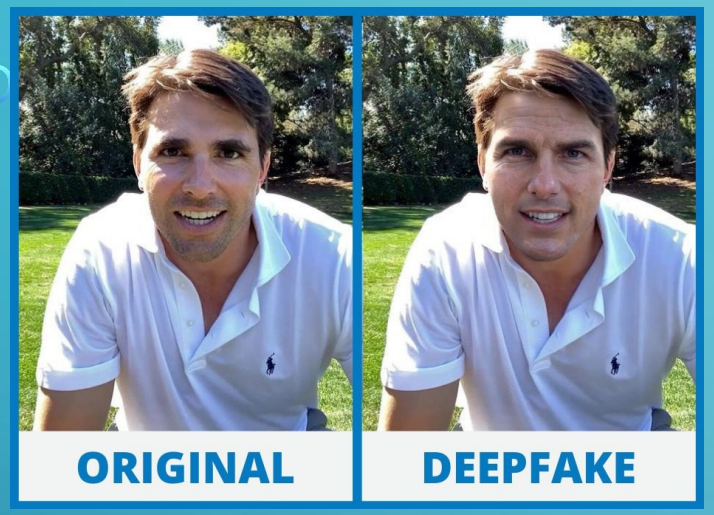


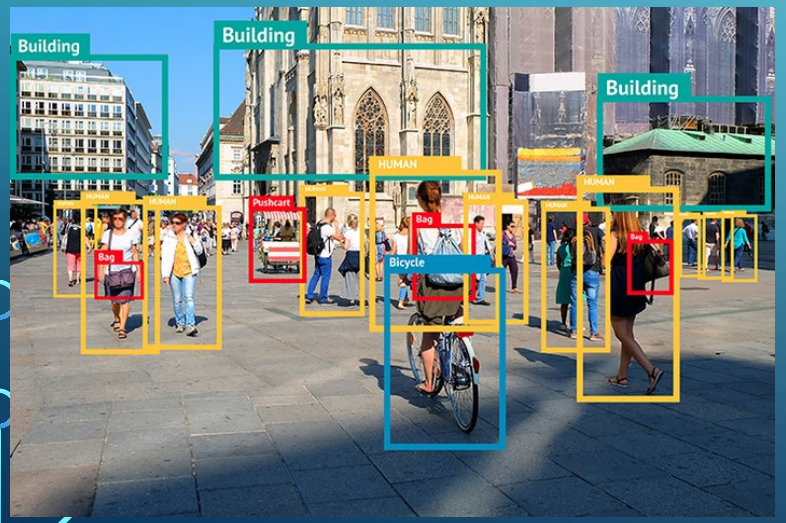
IMAGE PROCESSING LAB 1

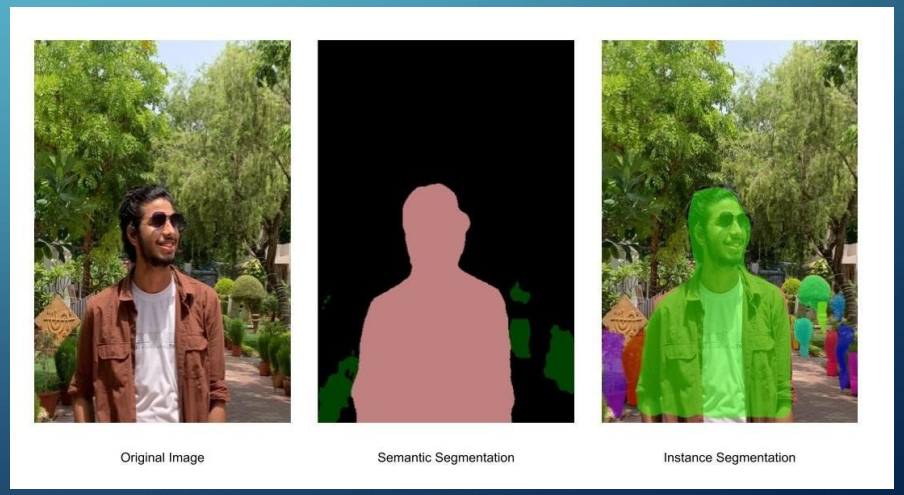
PRITAM MISHRA

Motivation


ORIGINAL
DEEPFAKE

Original
X-Pro II
Lomo-fi
Earlybird
Sutro
Toaster
Brannan
Inkwell
Walden
Hefe


Building
Building
Building
HUMAN
Pushcart
Bag
Bicycle


Original Image
Semantic Segmentation
Instance Segmentation

Lab Instructions

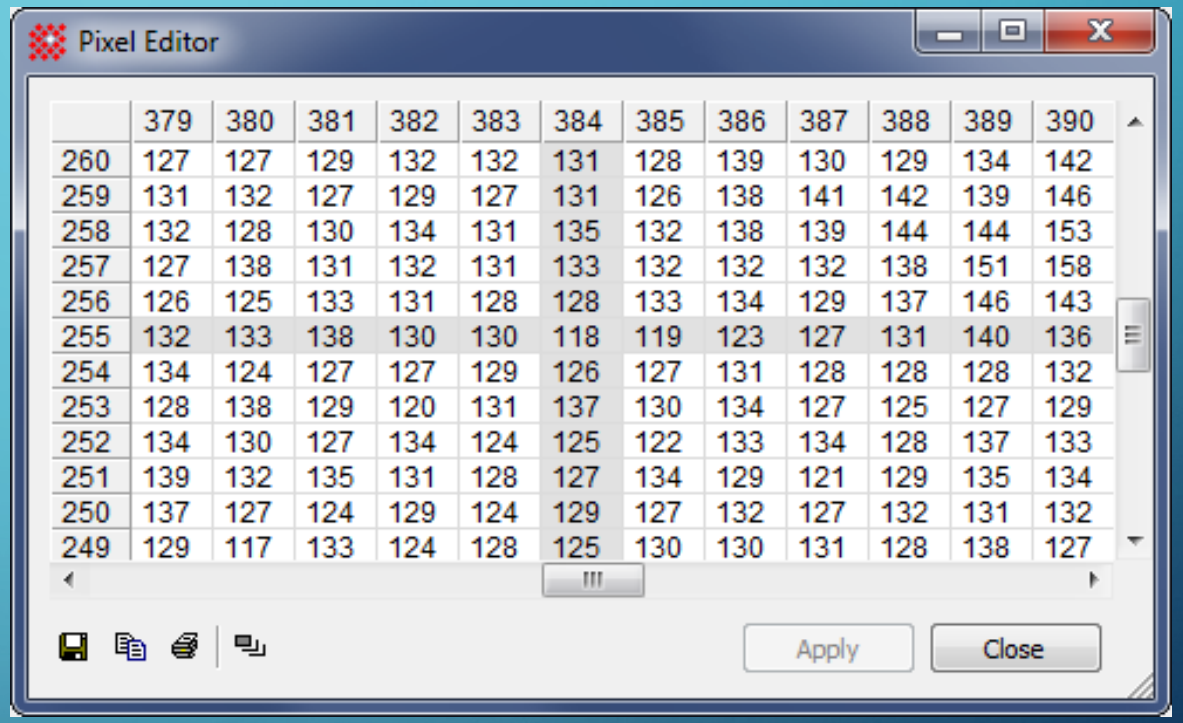
- We have 6 labs and 5 seminars in total.
- Usually, the deadline for each lab is before the start of next lab.
- You need to submit a report (detailed explanation) along with code before the deadline.
- 1 Viva Exam on the final Lab.
- Lab Marks ☾ 0.4 Average of [A] + 0.3 [B] + 0.3 [C]
- [A] ☾ All labs are delivered at its specific deadline.
- [B] ☾ Viva exam on the final lab (randomly one previous lab will be considered for viva for a group)
- [C] ☾ A question from the lab will appear in the midterm and final exams.
- Note ☾ [B] is mandatory to pass, If a group fails in [B], they need to retake the viva again.

What's new this year?

- Introduction of AI with Image Processing.
- All Labs will be in python, the most popular open-source language.
- Collaborative notebooks on Google Colab to run python code with free GPU.

What does an Image represent?

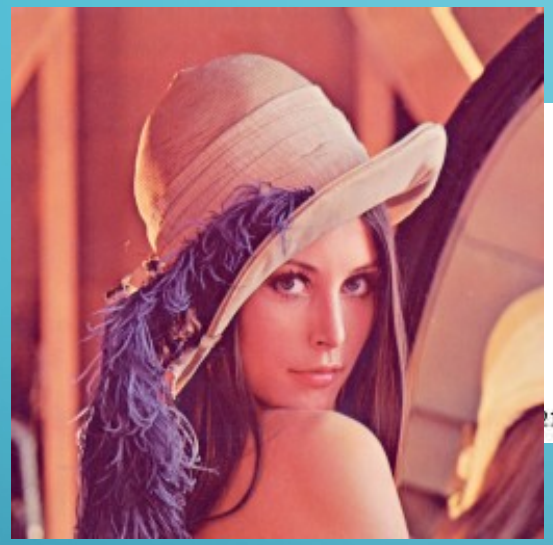
Image 512x512

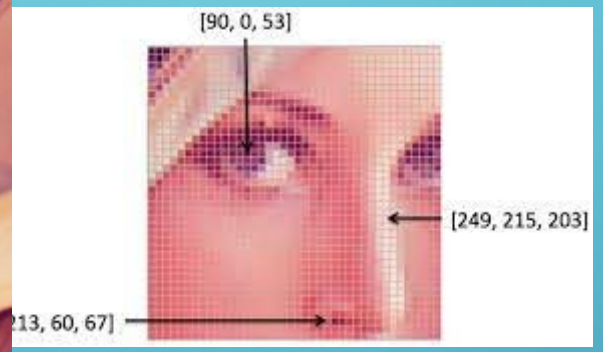


	379	380	381	382	383	384	385	386	387	388	389	390
260	127	127	129	132	132	131	128	139	130	129	134	142
259	131	132	127	129	127	131	126	138	141	142	139	146
258	132	128	130	134	131	135	132	138	139	144	144	153
257	127	138	131	132	131	133	132	132	132	138	151	158
256	126	125	133	131	128	128	133	134	129	137	146	143
255	132	133	138	130	130	118	119	123	127	131	140	136
254	134	124	127	127	129	126	127	131	128	128	128	132
253	128	138	129	120	131	137	130	134	127	125	127	129
252	134	130	127	134	124	125	122	133	134	128	137	133
251	139	132	135	131	128	127	134	129	121	129	135	134
250	137	127	124	129	124	129	127	132	127	132	131	132
249	129	117	133	124	128	125	130	130	131	128	138	127

Matrix 512x512 with values[0 , 255]

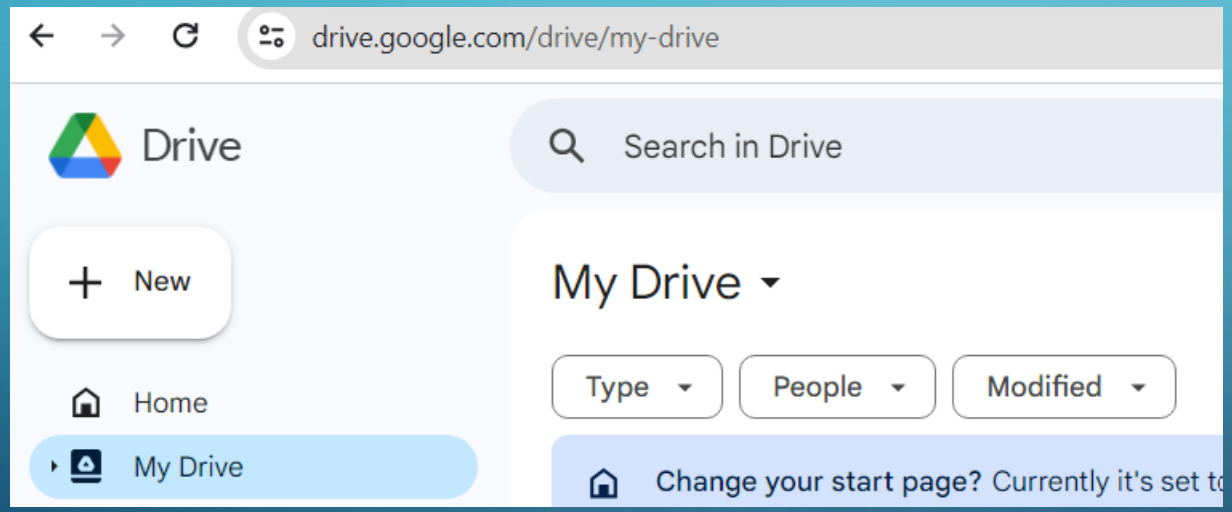
What does an Image represent?





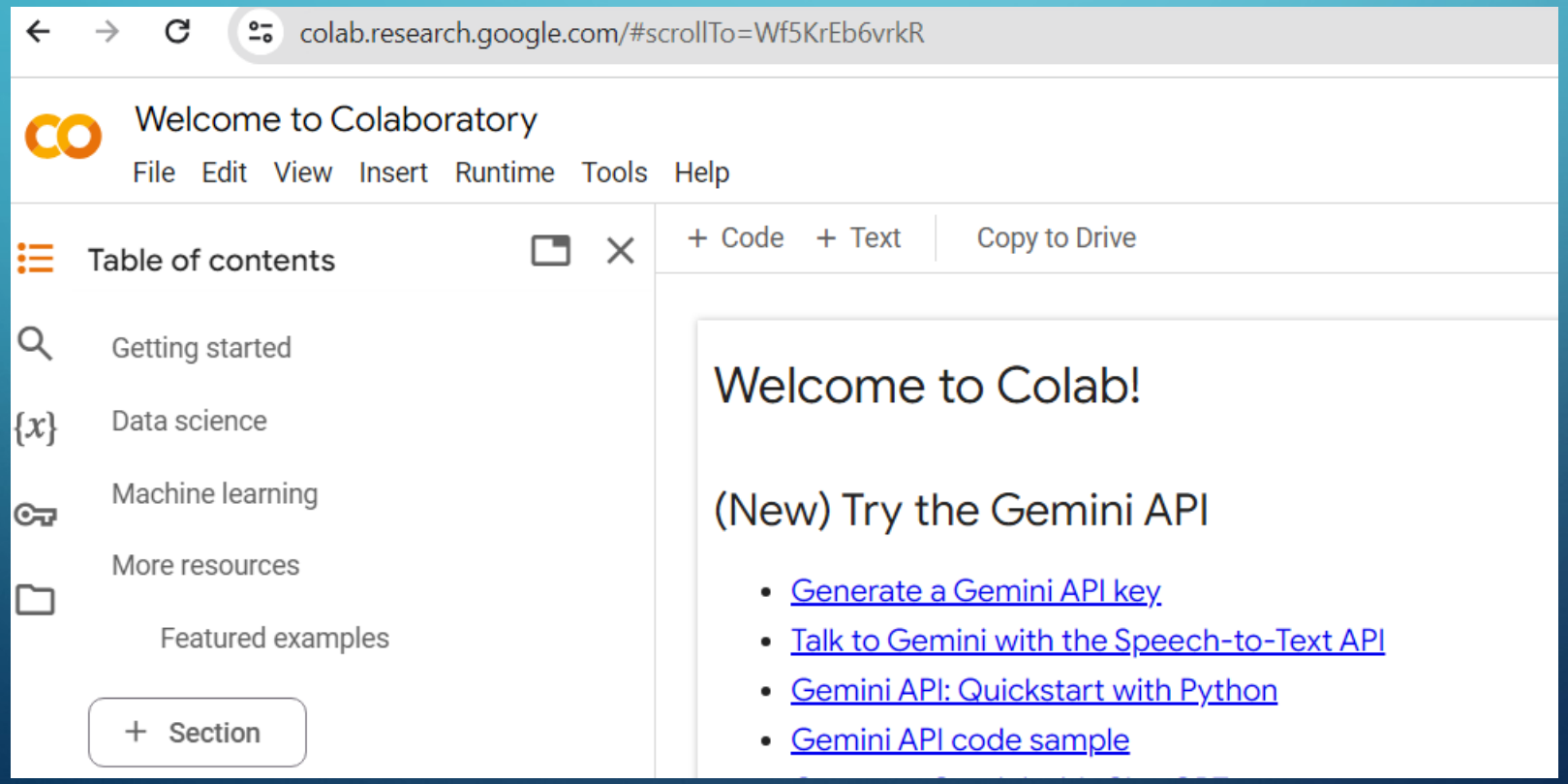
Working with Google Colab

- Go to your google drive and select “My Drive”.
- Choose new ☾ new folder.
- Select the Image Processing Lab 1 folder and upload it.



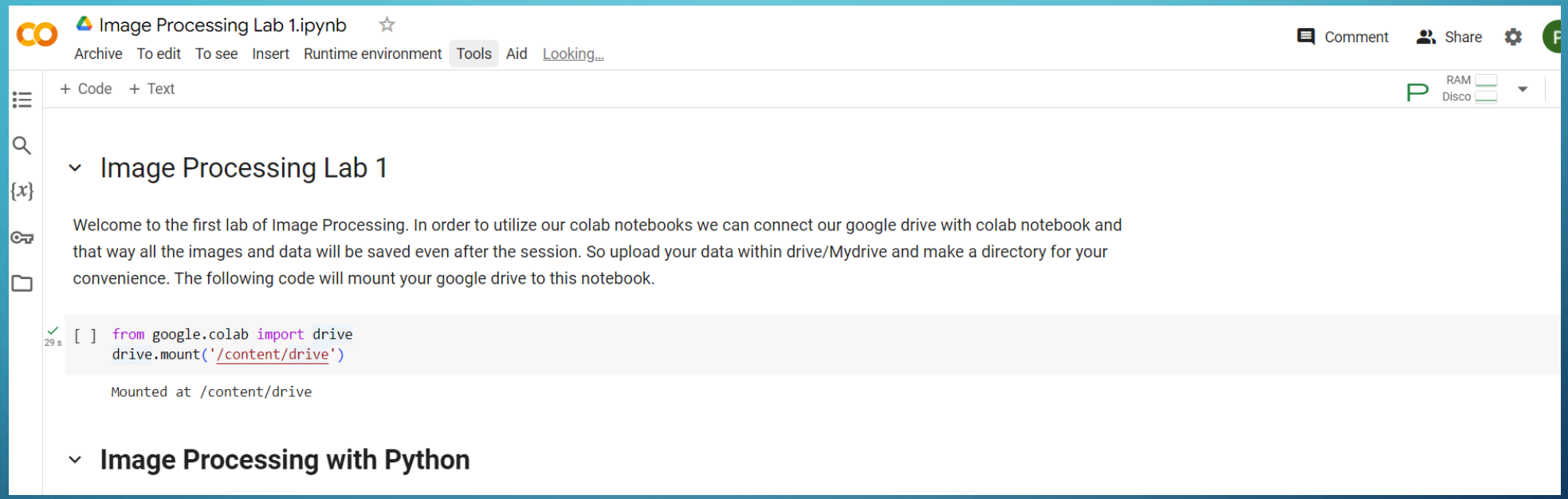
Working with Google Colab

- Now go to google Colab.
- Choose File ☾ upload notebook.
- Choose browse and select the file “Image Processing Lab 1.ipynb”.



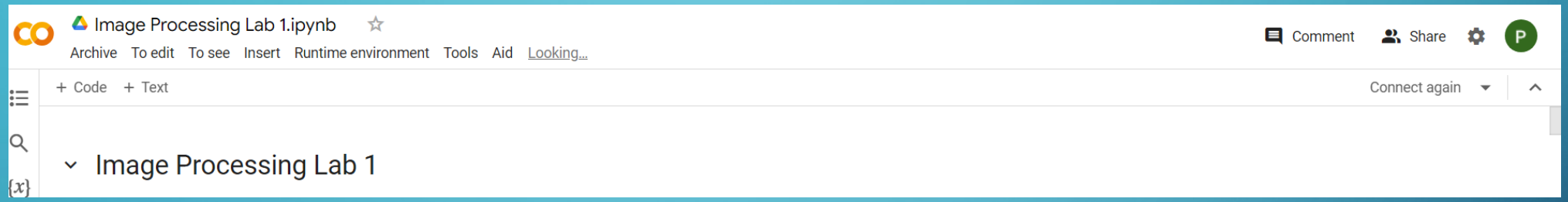
Working with Google Colab

- If it’s done correctly, the notebook will appear like this.

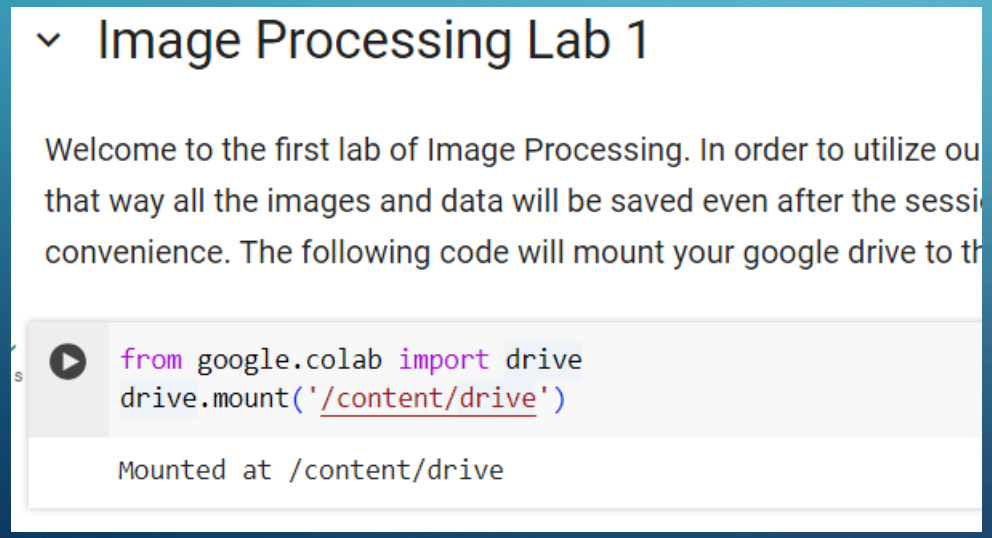


Working with Google Colab

- Let’s check if we can access the google drive files from here.

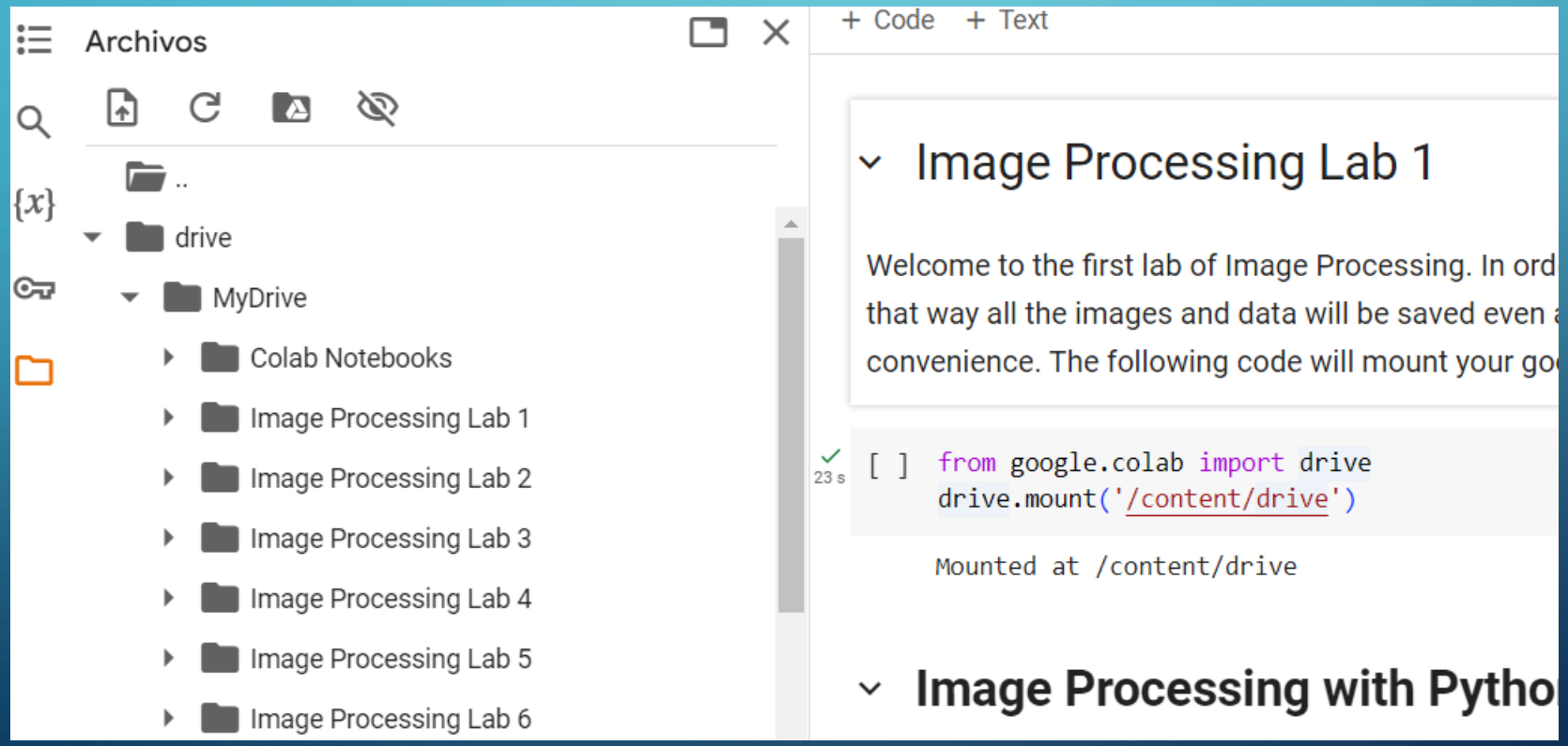


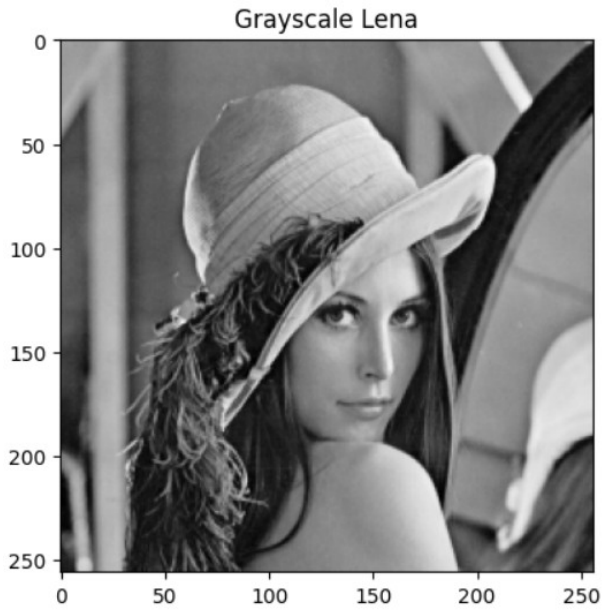
- Once connected run the first cell to mount your Google drive on Colab.



Working with Google Colab

- Now check the folder option on the left side.
- Select drive ☾ MyDrive – Now you can see the folder you uploaded here.






```
[ ] import matplotlib.pyplot as plt
    from PIL import Image
    import matplotlib.image as mpimg
    import numpy as np
    path = '/content/drive/MyDrive/Image Processing Lab 1/'
    lena = np.asarray(Image.open(path+"lena.png"))
```

```
[ ] def show_image(img, title):
      imgplot = plt.imshow(img, cmap='gray')
      plt.title(title)
      plt.show()
```

```
[ ] show_image(lena,"Grayscale Lena")
```

Let's check, for instance, the position (10,20) of a, which will indicate the value of the image pixel in the 10th row and 20th column.

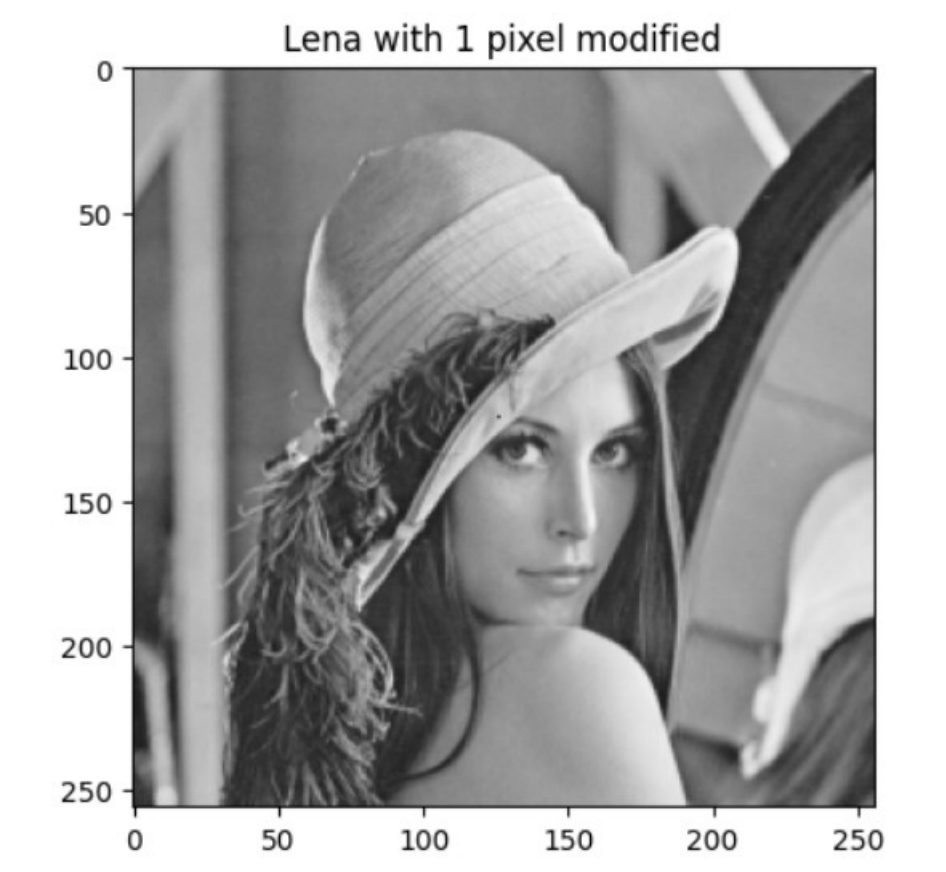
```
[ ] lena[10,20]
```

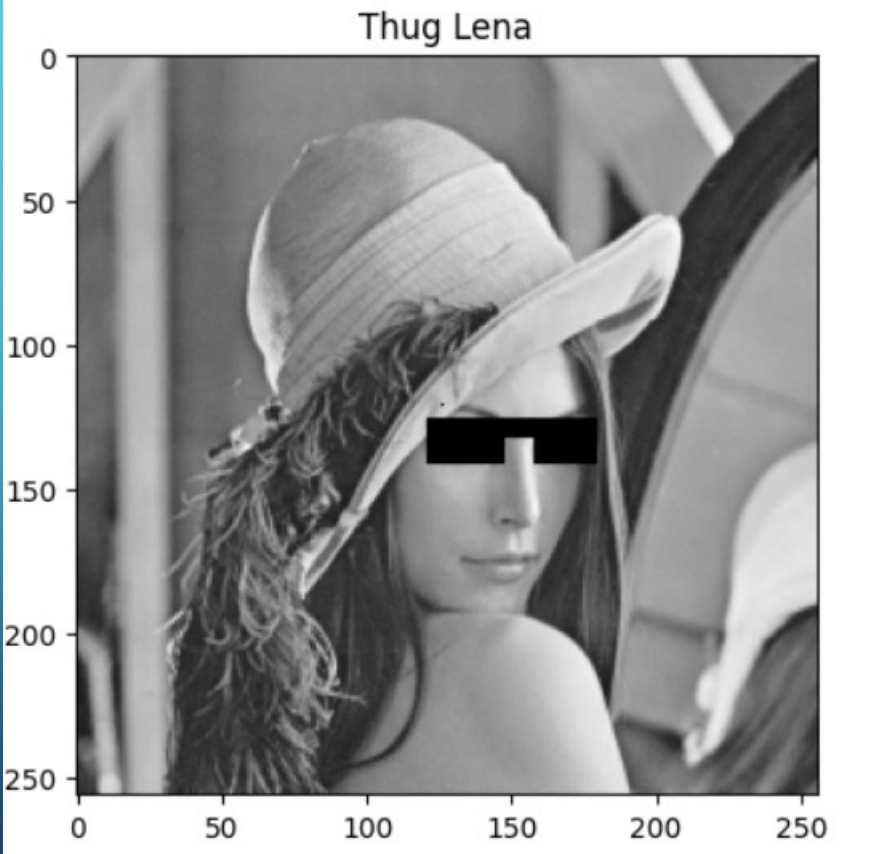
```
164
```

PLOTTING AN IMAGE IN PYTHON

```
[ ] lena[121,126] = 0
    show_image(lena, "Lena with 1 pixel modified")
```

```
[ ] lena[126:133,121:180] = 0;
    show_image(lena,'Almost Thug Lena')
```





Reading an Image in Double

```
lenaDouble = mpimg.imread(path+"lena.png")
show_image(lenaDouble,"Lena in Double Type")
```

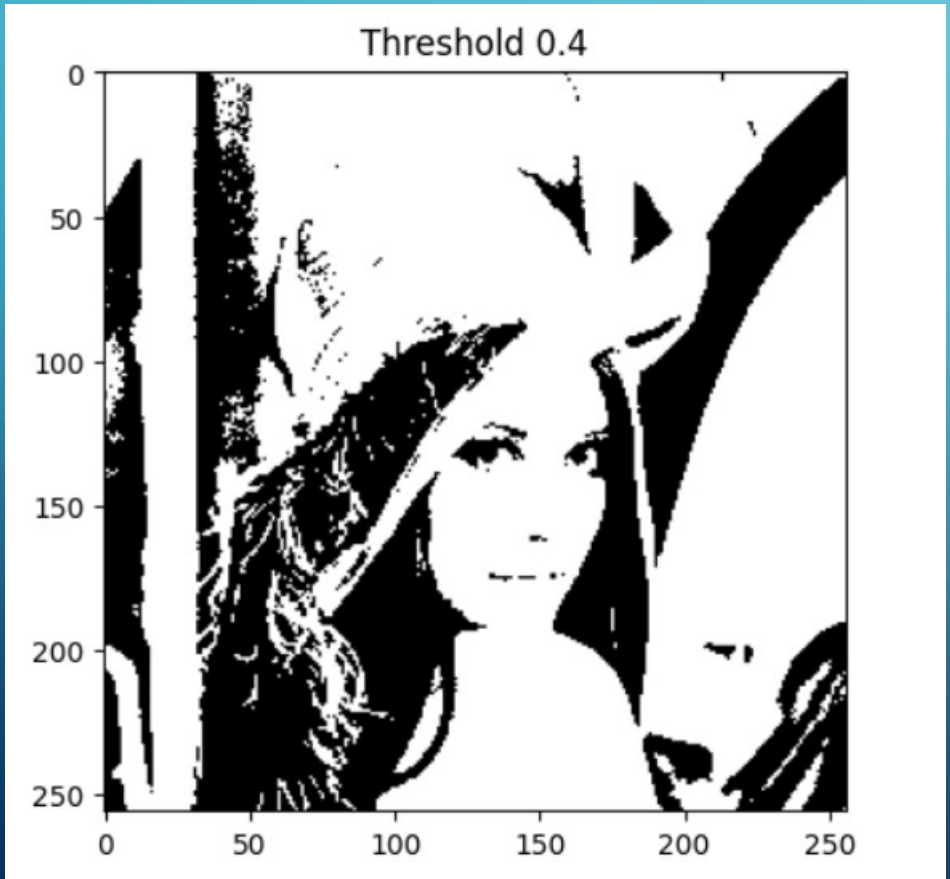
```
lena/255.0
```

```
lenaDouble
```

```
array([[0.63529414, 0.6313726 , 0.62352943, ..., 0.6627451 , 0.6666667 ,
        0.5529412 ],
       [0.63529414, 0.6313726 , 0.62352943, ..., 0.6627451 , 0.6666667 ,
        0.5529412 ],
       [0.63529414, 0.62352943, 0.62352943, ..., 0.61960787, 0.5921569 ,
        0.4509804 ],
       ...,
       [0.19607843, 0.19607843, 0.21176471, ..., 0.31764707, 0.3647059 ,
        0.36078432],
       [0.16862746, 0.1882353 , 0.19607843, ..., 0.36078432, 0.39607844,
        0.38039216],
       [0.16862746, 0.20392157, 0.19607843, ..., 0.39607844, 0.4       ,
        0.41568628]], dtype=float32)
```

Creating an Image Mask

```
[ ]  lenaBinTh = lenaDouble > 0.4
     show_image(lenaBinTh, "Threshold 0.4")
```




```
[ ]  lenaBinTh

     array([[ True,  True,  True, ...,  True,  True,  True],
            [ True,  True,  True, ...,  True,  True,  True],
            [ True,  True,  True, ...,  True,  True,  True],
            ...,
            [False, False, False, ..., False, False, False],
            [False, False, False, ..., False, False, False],
            [False, False, False, ..., False, False, True]])
```

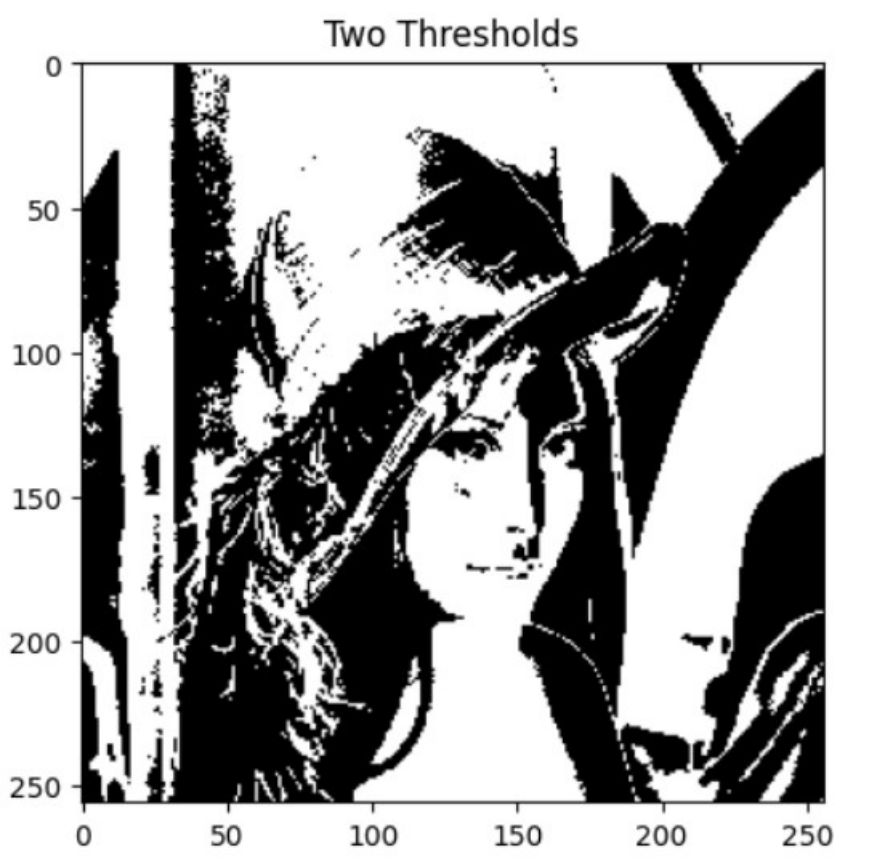
What would be the output of combining two masks?

```
[ ] lenaBinThB = lenaDouble < 0.7;
    lenaBinThB
```

```
array([[ True,  True,  True, ...,  True,  True,  True],
       [ True,  True,  True, ...,  True,  True,  True],
       [ True,  True,  True, ...,  True,  True,  True],
       ...,
       [ True,  True,  True, ...,  True,  True,  True],
       [ True,  True,  True, ...,  True,  True,  True],
       [ True,  True,  True, ...,  True,  True,  True]])
```

```
[ ] lenaBin2Th = lenaBinThB * lenaBinTh
    lenaBin2Th
```

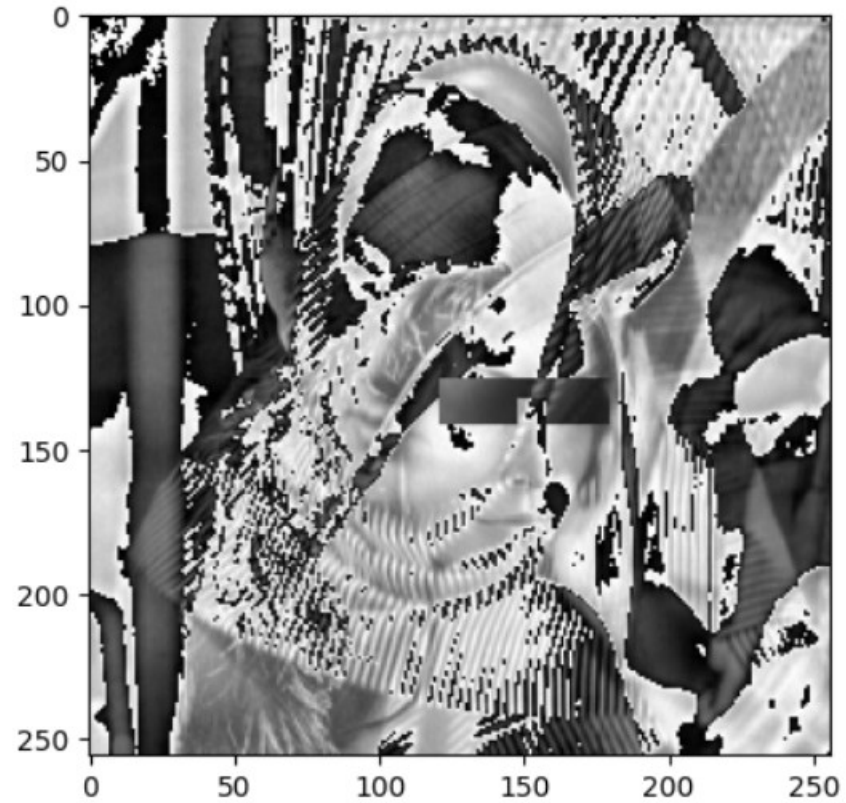
```
array([[ True,  True,  True, ...,  True,  True,  True],
       [ True,  True,  True, ...,  True,  True,  True],
       [ True,  True,  True, ...,  True,  True,  True],
       ...,
       [False, False, False, ..., False, False, False],
       [False, False, False, ..., False, False, False],
       [False, False, False, ..., False, False, True]])
```



What happens when we add two images?

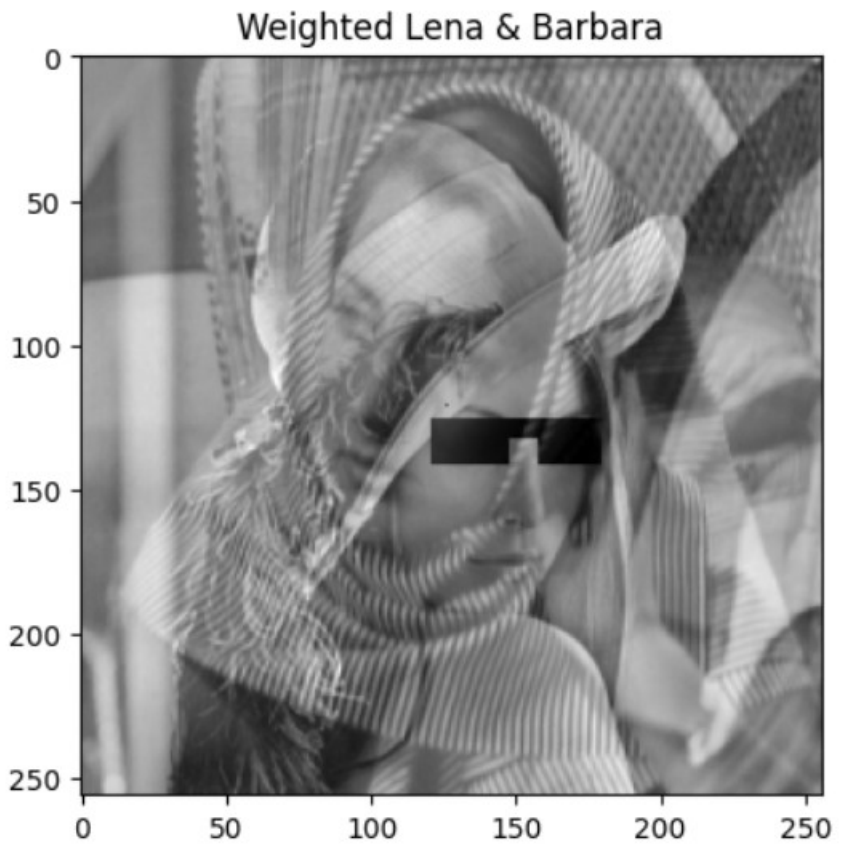
```
barbara = np.asarray(Image.open(path+"barbara.png"))
plt.imshow(lena_old+barbara, cmap='gray', vmin=0, vmax=255)
```

```
<matplotlib.image.AxesImage at 0x7f5b282603d0>
```



How can we ensure no image clipping?

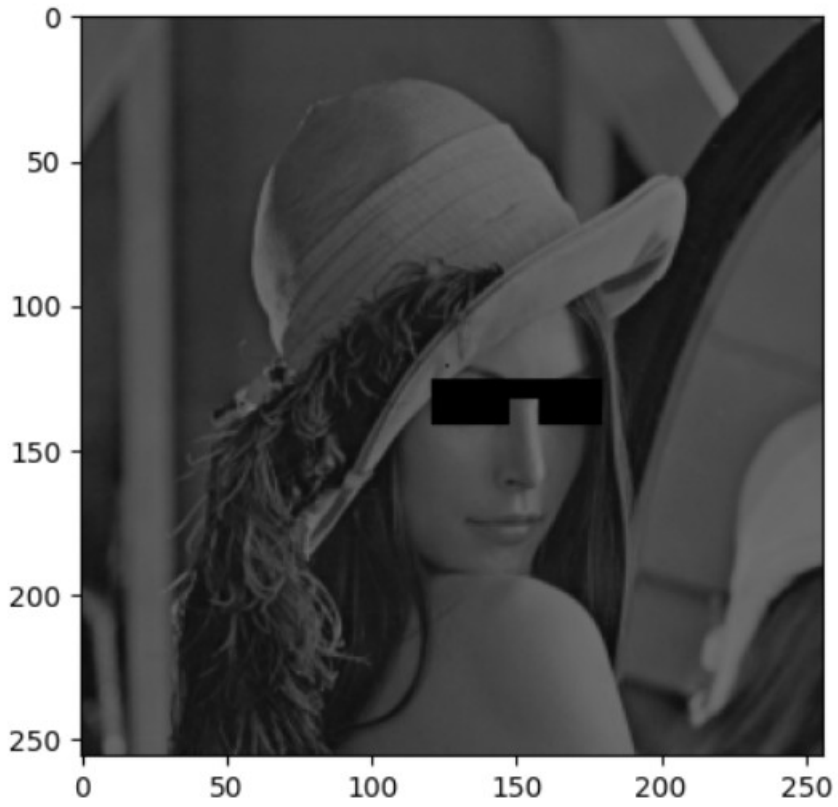
```
[ ] weightLena = 0.5; # Try to change it!
    weightBarbara = 1-weightLena;
    show_image(weightLena*lena_old + weightBarbara*barbara, "Weighted Lena & Barbara")
```



What happens if we multiply the Image by a constant?

```
#import cv2
constFac = 0.5; # change it and see what happens
mod_lena = constFac * lena
plt.imshow(mod_lena, cmap='gray', vmin=0, vmax=255)
```

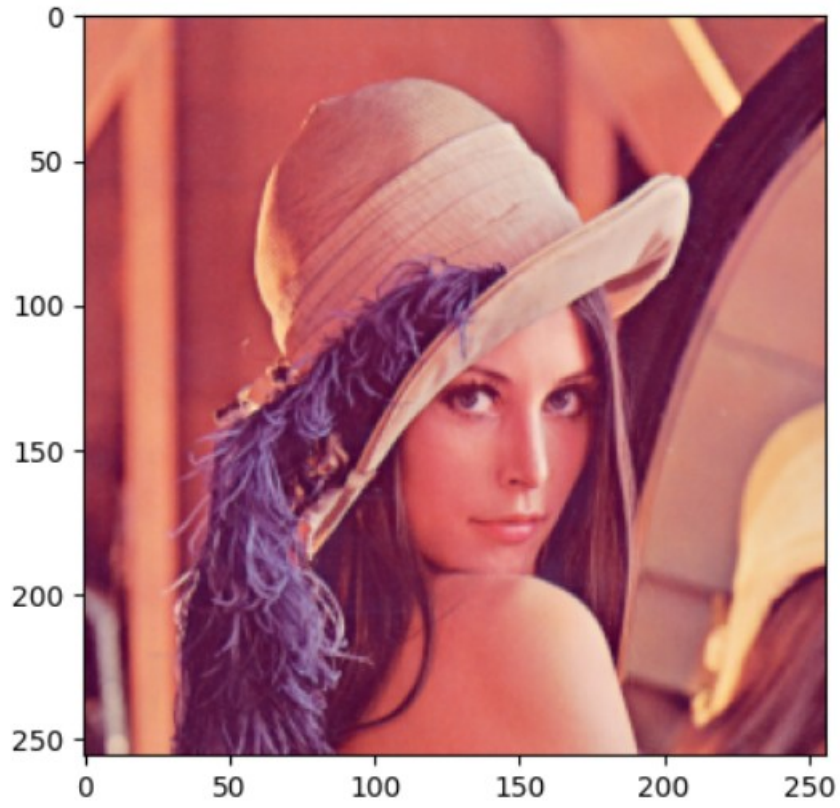
```
<matplotlib.image.AxesImage at 0x7f5b28197190>
```



Reading a RGB Image

```
lenacolor = mpimg.imread(path+"lena_color.png")
plt.imshow(lenacolor)
```

```
<matplotlib.image.AxesImage at 0x7f5b28251700>
```



How does the Numpy slicing work here?

```
r_lena = lenacolor[:,:,0];
g_lena = lenacolor[:,:,1];
b_lena = lenacolor[:,:,2];
```

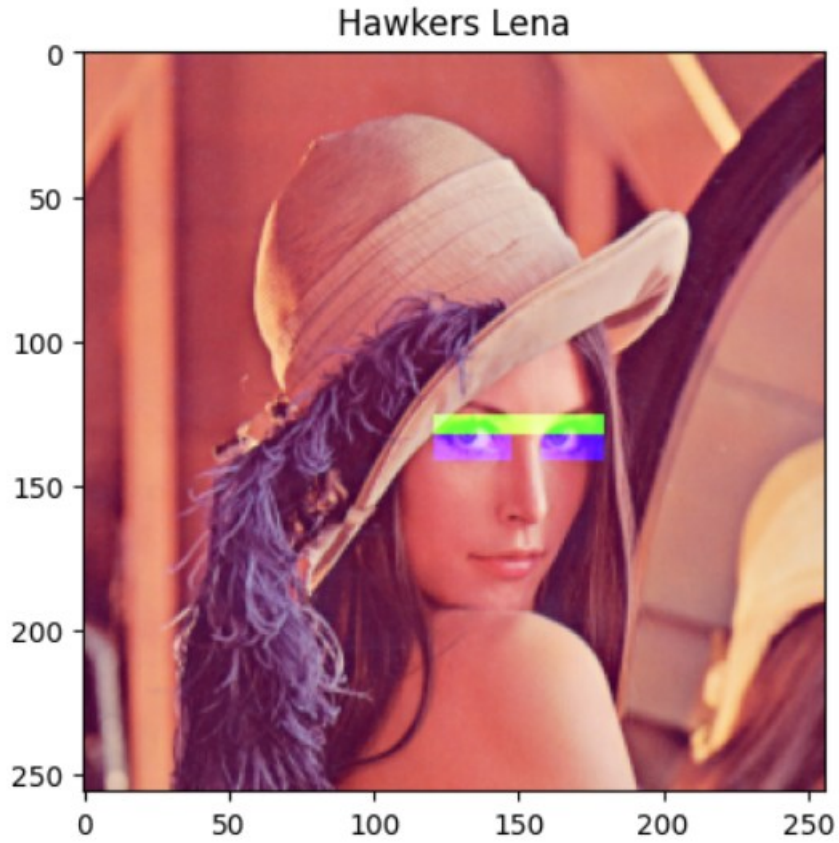
How would this image look like?

```
show_image(r_lena, "Red Channel")
```

How do we combine the image channels?

```
channels = []
g_lena[126:133,121:180] = 1; # Modify Green Channel
b_lena[133:142, 121:148] = 1; # Modify Blue Channel (1st part)
b_lena[133:142, 158:180] = 1; # Modify Blue Channel (2nd part)
lenaHawk = np.dstack((r_lena,g_lena,b_lena))
plt.title('Hawkers Lena')
plt.imshow(lenaHawk)
```

```
<matplotlib.image.AxesImage at 0x7f5b28100310>
```

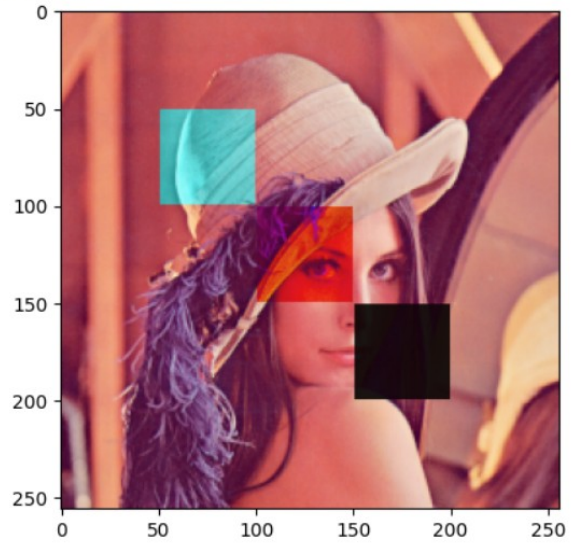


From RGB to HSV space

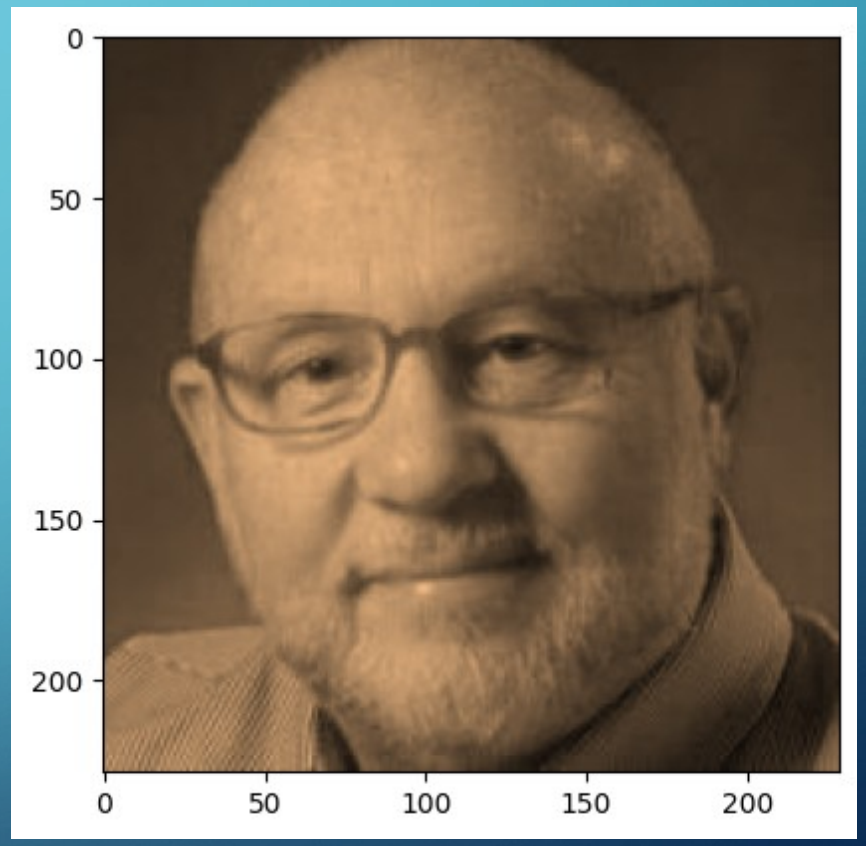
How would you define HSV?

```
#import cv2
#import colorsys
from skimage.color import rgb2hsv, hsv2rgb
lena_rgb = np.asarray(Image.open(path+"lena_color.png"))
lena_hsv = rgb2hsv(lena_rgb)
im_hue = lena_hsv[:,:,0]
im_sat = lena_hsv[:,:,1]
im_val = lena_hsv[:,:,2]
im_hue[51:100,51:100] = 0.5; # Set to a blue-ish tone
im_sat[101:150,101:150] = 0.9; # Saturate the color
im_val[151:200,151:200] = 0.1; # Set a low value (~ grayscale value)
im_hsv = np.dstack((im_hue,im_sat,im_val))
im_rgb = hsv2rgb(im_hsv)
plt.imshow(im_rgb)
```

```
<matplotlib.image.AxesImage at 0x7f5b04245790>
```

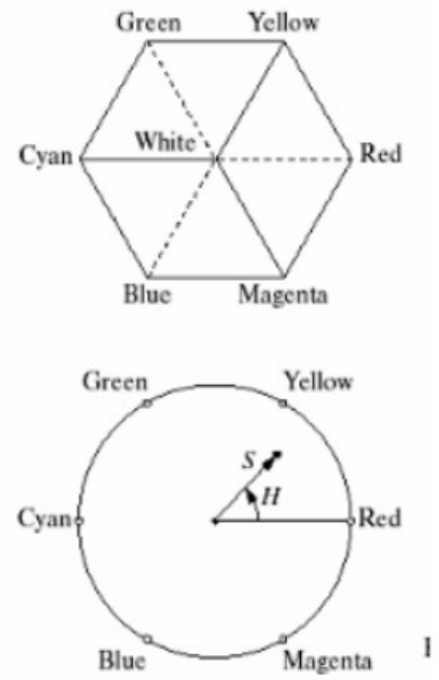


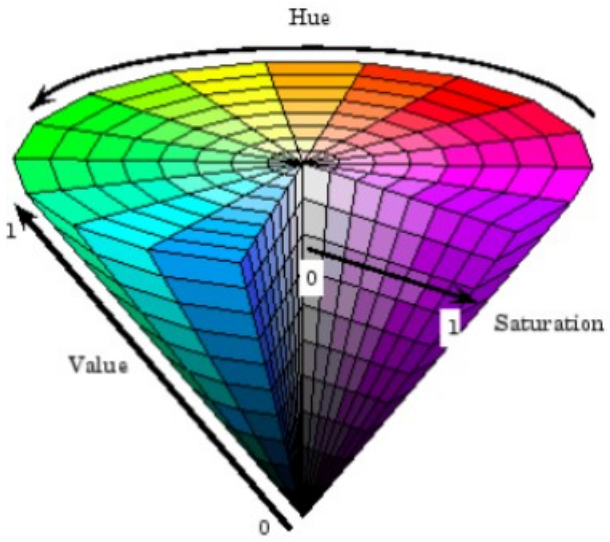
Exercise 1

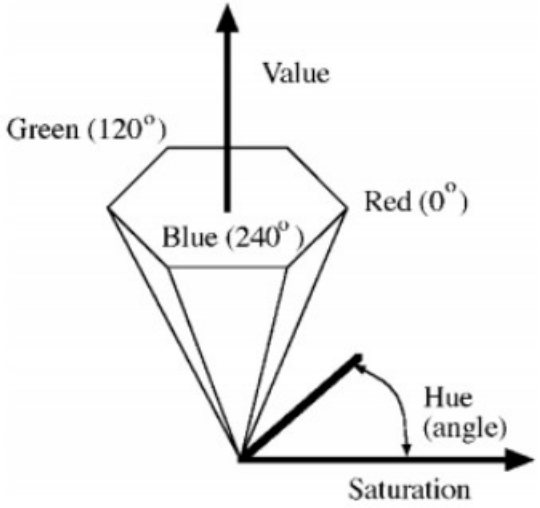


Hints: Exercise 1

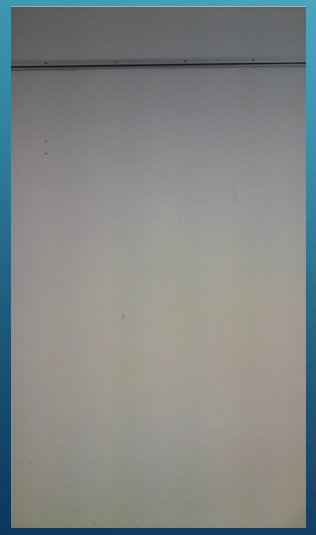
- Perceptually meaningful dimensions
- RGB cube on its vertex







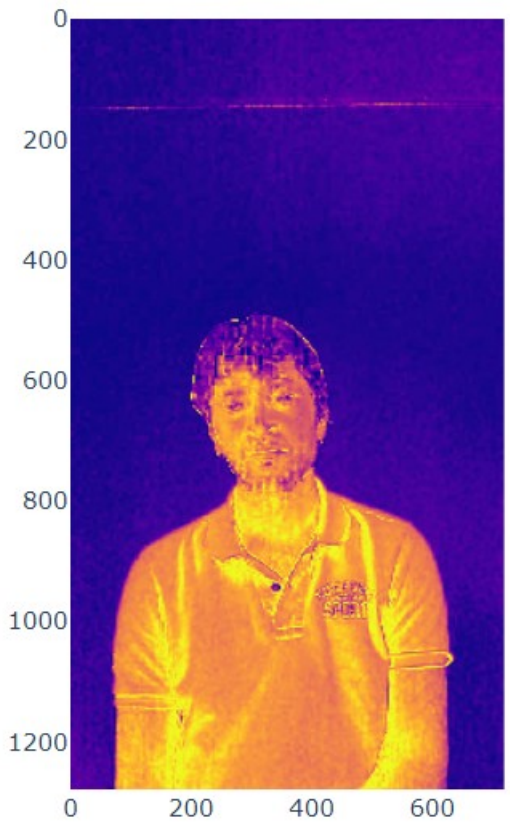
Exercise 2



An Important Tool to Consider

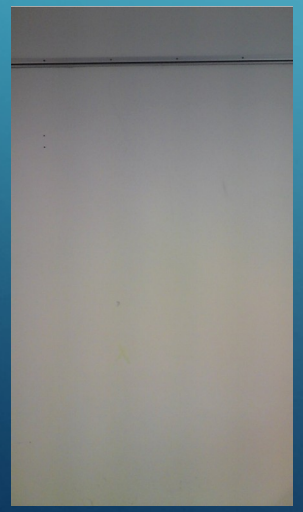
Practice Session

```
person = mpimg.imread(path+"FG1P.jpg")
person_hsv = rgb2hsv(person)
import plotly.express as px
px.imshow(person_hsv[:,:,1])
```



Hints: Exercise 2

Create a Mask of this guy without skin pixels.



Create a mask of backgroun d pixels that replace skin pixels.